Artificial Neural Networks (ANNs) are a key technology in modern **Artificial Intelligence** and **Machine Learning**. They are computational models inspired by the structure and functioning of the human brain. Just as the brain contains interconnected neurons that process information, ANNs consist of artificial neurons organized into layers that work together to recognize patterns and make predictions.

A typical neural network has three main parts: the input layer, hidden layers, and the output layer. The input layer receives raw data such as images, numbers, or text. Hidden layers process this information through mathematical operations and weighted connections. Finally, the output layer produces the result, such as a classification, prediction, or decision.

During training, the network learns from data by adjusting the weights of the connections between neurons. This process usually involves algorithms such as **Backpropagation**, which reduces errors between predicted outputs and actual results. With enough data and training, neural networks can perform complex tasks with high accuracy.

Artificial neural networks are widely used in applications like image recognition, speech recognition, recommendation systems, and medical diagnosis. Advances in computing power and large datasets have enabled deeper and more powerful models, often called **Deep Learning**, which continue to transform many industries and research fields.